

A floor plan, one query, two answers

walls-with-doorways are non-convex unions: the sampling-based generator's cuts leave accepted wedges at the jambs. The taut geometric optimum hugs the jambs and clips the wall; the physical layer reroutes through the doorways and certifies.

convex pillar: ONE separating-hyperplane cut, exact (leak = 0)

residual wedges the audit accepted (red, $\epsilon=0.01$) on faces and door jambs

start

goal

--- GCS* / every geometric GCS answer: cost 7.68 — CLIPS THE WALL
— ours (validate → repair → certify): cost 7.89 (+2.8%) — through the doorways, certified

